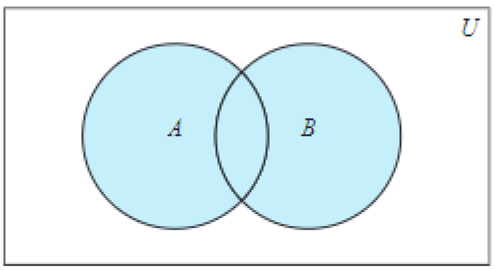
Note 2.2 Set Operations

Introduction

This section, we will learn the ways to combine the sets.

- **Definition of Union**
  Let A and B be sets. The union of the sets A and B, denoted by $A \cup B$ is the set that contains those elements that are either in A or in B, or in both. That is,

$$A \cup B = \{x | x \in A \vee x \in B\}$$

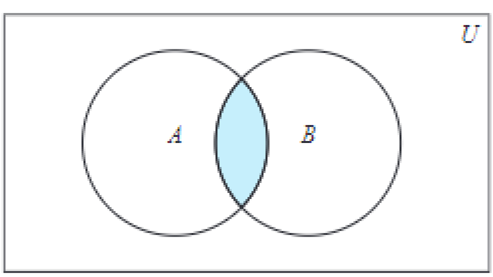


$A \cup B$ is shaded.

FIGURE 1 Venn Diagram of the Union of A and B.

- **Definition of Intersection**
  Let A and B be sets. The intersection of the sets A and B, denoted by $A \cap B$, is the set containing those elements in both A and B. That is,

$$A \cap B = \{x | x \in A \wedge x \in B\}$$



$A \cap B$ is shaded.

FIGURE 2 Venn Diagram of the Intersection of A and B.

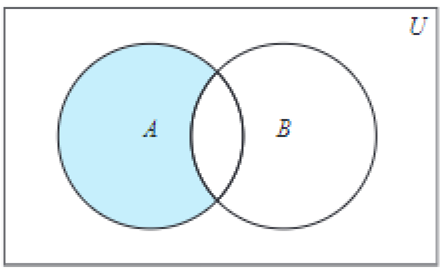
- **Definition of Disjoint**
  Two sets are called disjoint if their intersection is the empty set.

- **The Principle of Inclusion-Exclusion**
  For two sets A and B,

$$|A \cup B| = |A| + |B| - |A \cap B|$$

- **Definition of Difference**
  Let A and B be sets. The difference of A and B, denoted by $A - B$, is the set containing those elements that are in A but not in B. The difference of A and B is also called the complement of B with respect to A. That is,

$$A - B = \{x | x \in A \wedge x \notin B\}$$



$A - B$ is shaded.

FIGURE 3 **Venn Diagram for the Difference of A and B.**

- **Definition of Complement**
  Let U be the universal set. The complement of the set A, denoted by $\overline{A}$, is the complement of A with respect to U. Therefore, the complement of the set A is $U - A$. That is,

$$\overline{A} = \{x | x \in U \wedge x \notin A\}$$

  and there exists

$$A - B = A \cap \overline{B}$$

Set Identities

Table 1	Set Identities
Identity	**Name**
$A \cap U = A, A \cup \emptyset = A$	Identity laws
$A \cup U = U, A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A, A \cap A = A$	Idempotent laws
$\overline{(\overline{A})} = A$	Complementation law

Table 1	Set Identities
$A \cup B = B \cup A, A \cap B = B \cap A$	Commutative laws
$A \cup (B \cup C) = (A \cup B) \cup C, A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \overline{A} \cup \overline{B}, \overline{A \cup B} = \overline{A} \cap \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A, A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U, A \cap \overline{A} = \emptyset$	Complement laws

Generalized Unions and Intersections

- **Definition of Union for More Sets**
  The union of a collection of sets is the set that contains those elements that are members of at least one set in the collection. And we use

$$A_1 \cup A_2 \cup \cdots \cup A_n = \bigcup_{i=1}^{n} A_i$$

  to denote the union of the sets $A_1, A_2, \cdots, A_n$.
- **Definition of Intersection for More Sets**
  The intersection of a collection of sets is the set that contains those elements that are members of all the sets in the collection. And we use

$$A_1 \cap A_2 \cap \cdots \cap A_n = \bigcap_{i=1}^{n} A_i$$

  to denote the intersection of the sets $A_1, A_2, \cdots, A_n$.

Computer Representation of Sets

The sets stored in the computer use a method for storing elements using an arbitrary ordering of the elements of the universal set. Let the universal set U is finite. First, specify an arbitrary ordering of the elements of U, for instance $a_1, a_2, \cdots, a_n$. Then represent a subset A of U with the bit string of length n, where the ith bits in this string is 1 if a_i belongs to A and 0 if a_i does not belong to A.